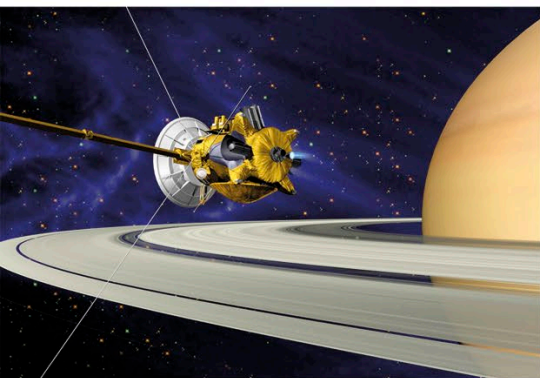


Orbiting Saturn

- **What?** Cassini/Huygens
- **Who?** NASA/ESA
- **Where and when?** US/Europe, launched 1997

Following its early flybys, NASA launched orbiter probes to the giant planets Jupiter and Saturn. The bus-sized Cassini reached Saturn in 2004. It spent more than a decade investigating the planet and its rings and moons. It also released a European-built lander called Huygens that touched down on Saturn's mysterious giant moon, Titan.



Close-up on Jupiter

- **What?** Juno
- **Who?** NASA
- **Where and when?** US, launched 2011

Juno is NASA's latest Jupiter orbiter. Part of NASA's New Frontiers program, it is designed to study the giant planet in

Illustration of Philae on Comet 67P



Landing on a comet

- **What?** Rosetta/Philae
- **Who?** ESA
- **Where and when?** Europe, launched 2004

The European Rosetta probe took more than a decade to reach its target, a comet known as 67P, before orbiting it for about two years. Shortly after arrival, Rosetta released a small lander called Philae, which unfortunately bounced into a deeply shadowed area where it could not charge its solar cells. However, at the end of the mission, Rosetta itself was steered to a successful landing.

Racing to Pluto

- **What?** New Horizons
- **Who?** NASA
- **Where and when?** US, launched 2006

Beyond the planets, the outer solar system is encircled by a ring of small icy objects, known as the Kuiper Belt. In 2006, NASA launched a high-speed probe to Pluto, one of the largest and closest of these worlds. It became the fastest object to leave Earth orbit, and after a slingshot (see p.284) at Jupiter, it flew past Pluto in July 2015, before continuing onward to new targets.



FAST FACTS

- Unlike satellites, space probes need sufficient speed to escape Earth's gravity entirely—launch must achieve an “escape velocity” of 7 miles per second (11.2 km/s).
- Voyagers 1 and 2 both carry a “golden record” containing greetings in various languages, music, birdsong, and other sounds from life on Earth for future spacefaring civilizations to play.

Bands of clouds and storms swirl across Jupiter's southern hemisphere in this view from Juno.

